

भारतीय मानक
Indian Standard

IS 14542 : 2025

आंशिक रूप से मलाईरहित दूध
पाउडर — विशिष्टि
(पहला पुनरीक्षण)

Partly Skimmed Milk Powder —
Specification
(First Revision)

ICS 67.100.10

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Price Group 6

Dairy Products Sectional Committee, FAD 19

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Dairy Products Sectional Committee had been approved by the Food and Agriculture Division Council.

The milk production in our country is characterized by seasonal variations and drying of milk, an important method of preservation, facilitates later consumption during the lean season. The dried milk products are an essential part of the chain between the producer and the consumer.

This standard was first published in 1998 subsequent to the amendment of Sub Rule 16 of Rule 49 of PFA Rules, 1955 vide Gazette Notification No. G.S.R. 41(E) dated 29 January 1997 issued by the Government of India, thereby, including partly skimmed milk powder, among other products, under mandatory BIS Certification. An amendment was later issued to IS 14542 : 1998 in November 2012.

This revision has been brought out to harmonize the standard with the *Food Safety and Standards (Food Products Standards and Food Additives) Regulations*, 2011. In this revision, the following major changes have been done:

- a) Description of the product has been revised;
- b) Raw materials permitted for protein adjustment have been specified;
- c) Specification of Type II partly skimmed milk powder (sour) has been removed;
- d) Requirement of milk protein in milk solids not fat has been incorporated in place of total solids;
- e) Ash content expression has been changed to moisture and fat free basis instead of dry basis;
- f) *Food Safety and Standards (Food Products Standards and Food Additives) Regulations*, 2011 have been referred regarding the use of food additives;
- g) Microbiological limits have been aligned with the *Food Safety and Standards (Food Products Standards and Food Additives) Regulations*, 2011 and latest methods for microbiological analysis have been referred. Additionally, limit of *E. coli* has been specified;
- h) *Food Safety and Standards (Contaminants, Toxins and Residues) Regulations*, 2011 have been referred for the limits of pesticide residues, antibiotic and veterinary drug residues; and
- i) Limits of heavy metals and other contaminants or toxic substances (aflatoxin M₁ and melamine) have been aligned with FSSAI Regulations.

Separate Indian Standards exist for whole milk powder (IS 1165 : 2022) and skimmed milk powder [IS 13334 (Part 1) : 2014 'Skimmed milk powder — Specification: Part 1 Standard grade (*second revision*)' and IS 13334 (Part 2) : 2014 Skimmed milk powder — Specification: Part 2 Extra grade (*first revision*)].

While formulating this standard, due consideration has been given to the *Food Safety and Standards (Food Products Standards and Food Additives) Regulations*, 2011 and *Legal Metrology (Packaged Commodities) Rules*, 2011. However, this standard is subject to restrictions imposed under these rules, wherever applicable.

The composition of the Committee responsible for revision of the standard is given in [Annex C](#).

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

PARTLY SKIMMED MILK POWDER — SPECIFICATION

(*First Revision*)

1 SCOPE

This standard prescribes the requirements, methods of sampling and test for partly skimmed milk powder.

2 REFERENCES

The standards listed in [Annex A](#) contain provisions which through reference in this text, constitute provision of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards.

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Partly Skimmed Milk Powder — The product obtained from milk (for definition and type of milk, see 3 and 4 of IS 13688) by partial removal of water through spray drying or by roller drying.

4 REQUIREMENTS

4.1 Description

The product shall be white or white with a greenish tinge to light cream in colour. It shall be free from lumps except those that break up readily under slight pressure.

4.2 The product shall be free from extraneous matter, added colours and flavours.

4.3 The product may contain only permitted food additives within the limits as specified under the *Food Safety and Standards (Food Products Standards and Food Additives) Regulations, 2011*.

4.4 The flavour of the product or of the reconstituted milk shall be pleasant and clean. It shall be free from off flavours. It may have slightly cooked but not the burnt flavour. It is recommended that the flavour and taste may be judged on the basis of their sensory characteristics (see IS 10030).

4.5 The product shall also conform to the requirements given in [Table 1](#) and [Table 2](#).

4.6 The fat or protein content, or both of the milk may be adjusted, only to comply with the compositional requirements stated in [Table 1](#) by addition or withdrawal of milk constituents in such a way as not to alter the whey protein to casein ratio of the milk being adjusted. The following milk products are allowed for protein adjustment purposes.

4.6.1 Milk Retentate

The product obtained by concentrating milk protein by ultrafiltration of milk, partly skimmed milk, or skimmed milk.

4.7 Total sodium content in the product shall not be more than 650 mg/100 g solid-not-fat, when tested as per method prescribed in IS 12760.

NOTE — The maximum level does not apply to sodium that could be present due to the use of sodium containing additives in milk powders.

4.8 The pesticide residues, antibiotic and veterinary drug residues, if any, in the product shall not exceed the limits as prescribed in the *Food Safety and Standards (Contaminants, Toxins and Residues) Regulations, 2011*.

4.9 The metal contaminants and other contaminants or toxic substances (aflatoxin M₁ and melamine), if any, in the product shall not exceed the limits specified in [Table 3](#).

Table 1 Requirements for Partly Skimmed Milk Powder

(Clauses [4.5](#) and [4.6](#))

Sl No.	Characteristics	Requirements	Method of Test, Ref to
(1)	(2)	(3)	(4)
i)	Moisture, percent by mass, <i>Max</i>	4.0	IS 11623* or IS 16072 or ISO 21543 ¹⁾

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Table 1 (Concluded)

Sl No.	Characteristics	Requirements	Method of Test, Ref to
(1)	(2)	(3)	(4)
ii)	Milk protein in milk solids not fat, percent by mass, <i>Min</i>	34.0	IS 11917 ²⁾
iii)	Milk fat, percent by mass:		IS 19396 or Annex B of IS 13334 (Part 1)
	a) More than	1.5	
	b) Less than	26.0	
iv)	Insolubility index, ml, <i>Max</i> :		IS 12759
	a) Roller dried	15.0	
	b) Spray dried	1.5	
v)	Total ash (on moisture and fat free basis) percent by mass, <i>Max</i>	8.7	Annex C of IS 14433
vi)	Titrateable acidity, ml of 0.1 N NaOH/10 g of solids not fat	12 to18	IS 11766
vii)	Scorched particles, mg, <i>Max</i>	15.0 (equivalent to Disc B)	IS 13500

NOTE — In case of dispute, the method indicated by '**' shall be the referee method.

4.10 Hygienic Conditions

The material shall be manufactured and packed under hygienic conditions in licensed premises (see IS 2491). The basic principles of hygiene should be applied with appropriate modifications.

5 PACKING AND MARKING

5.1 Packing

5.1.1 Retail Packing

The product shall be packed in clean and sound metal containers (see IS 18427) or in a food grade flexible pack made from a film or combination of any of the substrates made of board, paper, polyethylene, polyester metalized film or aluminium foil in such a way so as to protect it from deterioration. In the case of the plastic material, only food grade plastic (see IS 10171) shall be used.

5.1.1.1 The product shall be packed in quantities as stipulated under the provisions of the *Legal Metrology (Packaged Commodities) Rules, 2011*.

5.1.1.2 Further encasing of individual retail packs may be carried out in bags/cartons of adequate strength as outlined in [5.1.2](#).

5.1.2 Bulk Packing

The product may be packed in quantities of 25 kg in bags of food grade polyethylene (see IS 10171) of minimum thickness 0.05 mm. The bags should be properly closed by tying with a string or a rubber band or heat sealed and shall be subsequently encased in multi-walled kraft paper, such as crepe kraft paper bags of not less than 80 g/m² (GSM) grade, appropriately lined and having two or more inner layers of plain kraft paper of not less than 80 g/m² (GSM) grade.

5.2 Marking

5.2.1 The package shall bear legibly and indelibly the following information:

- Name and grade of the material and brand name, if any;
- Name and address of the manufacturer;
- Batch or Code No.;
- Process of drying;
- Fat content;
- Date of manufacturing or packing;
- Net quantity (see [5.1.1.1](#));

¹⁾ Before being used, the instrument should be calibrated with adequate data of similar milk products and validated against reference standard method as per the procedure prescribed ISO 21543.

²⁾ Estimate milk protein content by the method prescribed in IS 11917. Calculate milk solid not fat percentage by subtracting milk fat and moisture from 100. Calculate milk protein in milk solid not fat, percent by mass as follows:

a) Milk protein in milk solid not fat, percent by mass = Percent milk protein × 100 Percent milk solids not fat content.

- h) Direction for storage;
- j) Expiry/use by date
- k) Direction for reconstitution;
- m) The container contains equivalent oflitre of skimmed milk;
- n) In the case of flexible pack, the following information shall be marked on the label: 'Once opened, the entire product content should immediately be placed in a clean air-tight container'; and
- p) Any other requirements under the *Legal Metrology (Packaged Commodities) Rules, 2011* and the *Food Safety and Standards (Labelling and Display) Regulations, 2020*.

5.2.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

6 SAMPLING

The method of drawing representative samples of the material and criteria for conformity shall be as prescribed in Annex C of IS 1165. To decide conformity of a lot to microbiological requirements, sampling shall be done as prescribed in [Table 2](#).

Table 2 Microbiological Requirements for Partly Skimmed Milk Powder

(Clauses [4.5](#) and [6](#))

SI No.	Characteristic	Requirement				Method of Test, Ref to
		Sampling Plan		Limit (cfu)		
		n	c	m	M	
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Aerobic plate count	5	2	$3 \times 10^4/\text{g}$	$5 \times 10^4/\text{g}$	IS 5402 (Part 1)
ii)	Coliform count	5	2	10/g	50/g	IS 5401 (Part 1)
iii)	<i>Staphylococcus aureus</i> (coagulase positive)	5	2	10/g	$1 \times 10^2/\text{g}$	IS 5887 (Part 8/ Sec 1* or 2)
iv)	Yeast and mould count	5	0	50/g	–	IS 16069 (Part 2)
v)	<i>Salmonella spp.</i>	5	0	Absent/ 25 g	–	IS 5887 (Part 3/ Sec 1)
vi)	<i>Listeria monocytogenes</i>	5	0	Absent/g	–	IS 14988 (Part 1)
vii)	<i>Bacillus cereus</i>	5	3	$5 \times 10^2/\text{g}$	$1 \times 10^3/\text{g}$	IS 5887 (Part 6)
viii)	<i>Sulphite reducing clostridia</i>	5	3	50/g	$1 \times 10^2/\text{g}$	IS 18349 (Part 1)
ix)	<i>E. coli</i>	5	0	Absent/g	–	IS 16424

NOTES

1 For sampling plan, see [Annex B](#).

2 In case of dispute, the method indicated by '*' shall be the referee method.

3 The requirement for *Salmonella* shall be tested in a laboratory situated away from the production area.

Table 3 Limits of Metal Contaminants and Naturally Occurring Toxic Substances

([Clause 4.9](#))

Sl No.	Contaminant/Toxin	Limit	Method of Test, Ref to
(1)	(2)	(3)	(4)
i)	Lead, ppm, <i>Max</i>	$0.02 \times \text{concentration factor}$	IS 12074
ii)	Arsenic, ppm, <i>Max</i> ¹⁾	$0.1 \times \text{concentration factor}$	IS 11124

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Table 3 (Concluded)

SI No. (1)	Contaminant/Toxin (2)	Limit (3)	Method of Test, Ref to (4)
iii)	Aflatoxin M ₁ , ppb, <i>Max</i>	6.0	IS 18108 or IS 18110*
iv)	Melamine, ppm, <i>Max</i>	2.5	IS 16195
NOTES			
1 Concentration factor shall be as communicated by the manufacturer. General guidance for calculation of concentration factor for milk powder is given below:			
$\text{Concentration factor} = \frac{\text{Total solids in milk powder}}{\text{Total solids in source milk}}$			
2 In case of dispute, the method indicated by '*' shall be the referee method.			

¹⁾ A maximum limit of 1.1 ppm shall be applicable for arsenic content in case the specified value (0.1 × concentration factor) goes beyond 1.1 ppm.

ANNEX A

(Clause 2)

LIST OF REFERRED STANDARDS

IS No.	Title	IS No.	Title
IS 1165 : 2022	Whole milk powder — Specification (sixth revision)	(Sec 2) : 2023/ISO 6888-2 : 2021	Method using rabbit plasma fibrinogen agar medium (first revision)
IS 2491 : 2024	Food hygiene — General principles — Code of practice (fourth revision)	IS 10030 : 1981	Method for sensory evaluation of milk powder
IS 5401 (Part 1) : 2012/ISO 4832 : 2006	Microbiology of food and animal feeding stuffs — Horizontal method for the detection and enumeration of coliforms: Part 1 Colony-count technique (second revision)	IS 10171 : 1999	Guide on suitability of plastics for food packaging (second revision)
IS 5402 (Part 1) : 2021/ISO 4833-1 : 2013	Microbiology of the food chain — Horizontal method for the enumeration of microorganisms: Part 1 Colony count at 30°C by the pour plate technique (third revision)	IS 11124 : 1984	Method for atomic absorption spectrophotometric determination of arsenic
IS 5887 (Part 3/Sec 1) : 2020/ISO 6579-1 : 2017	Methods for detection of bacteria responsible for food poisoning: Part 3 Horizontal method for the detection, enumeration and serotyping of <i>Salmonella</i> , Section 1 Detection of <i>Salmonella</i> spp. (third revision)	IS 11546 : 2012/ISO 707 : 2008	Milk and milk products — Guidance on sampling (second revision)
IS 5887 (Part 6) : 2012/ISO 7932 : 2004	Microbiology of food and animal feeding stuffs — Horizontal Method for the enumeration of presumptive <i>Bacillus cereus</i> : Part 6 Colony-count technique at 30°C (first revision)	IS 11623 : 2025/ISO 5537 : 2023	Dried milk and dried milk products — Determination of moisture content (reference method) (second revision)
IS 5887	Methods for detection of bacteria responsible for food poisoning:	IS 11766 : 1986/ISO 6092 : 1980	Method for determination of titratable acidity in milk powder and similar products (routine method)
(Part 8)	Horizontal method for the enumeration of coagulase-positive <i>Staphylococcus aureus</i> and other species),	IS 11917 : 2018/ISO 8968-1 : 2014	Milk and milk products — Determination of nitrogen content — Kjeldahl principle and crude protein calculation (first revision)
(Sec 1) : 2023/ISO 6888-1 : 2021	Method using Baird-Parker agar medium (first revision)	IS 12074 : 1987	Method for determination of lead by atomic absorption spectrophotometer
		IS 12759 : 2019	Dried milk and dried milk products — Determination of insolubility index (first revision)
		IS 12760 : 2012/ISO 8070 : 2007	Milk and milk products — Determination of calcium, sodium, potassium and magnesium contents — Atomic absorption spectrometric method (first revision)

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IS No.	Title	IS No.	Title
IS 13334 (Part 1) : 2025	Skimmed milk powder — Specification: Part 1 Standard grade (<i>third revision</i>)		Horizontal method for the detection and enumeration of presumptive <i>Escherichia coli</i> — Most Probable Number technique
IS 13500 : 1992	Spray dried milk powders — Scorched particles — Determination	IS 18108 : 2023/ ISO 14674 : 2005	Milk and milk powder — Determination of aflatoxin M ₁ content — Clean-up by immunoaffinity chromatography and determination by thin-layer chromatography
IS 13688 : 2020	Packaged pasteurized milk — Specification (<i>second revision</i>)		
IS 14433 : 2022	Infant milk substitutes — Specification (<i>second revision</i>)	IS 18110 : 2023/ ISO 14501 : 2021	Milk and milk powder — Determination of aflatoxin M ₁ content — Clean-up by immunoaffinity chromatography and determination by high-performance liquid chromatography
IS 14988 (Part 1) : 2020/ISO 11290-1 : 2017	Microbiology of the food chain — Horizontal method for detection and enumeration of <i>Listeria monocytogenes</i> and of <i>Listeria</i> spp.: Part 1 Detection method (<i>first revision</i>)	IS 18349 (Part 1) : 2023/ISO 15213-1 : 2023	Microbiology of the food chain — Horizontal method for the detection and enumeration of <i>Clostridium</i> spp.: Part 1 Enumeration of sulfite-reducing <i>Clostridium</i> spp. by colony-count technique
IS 16069 (Part 2) : 2013/ISO 21527-2 : 2008	Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of yeasts and moulds: Part 2 Colony count technique in products with water activity less than or equal to 0.95	IS 18427 : 2024	Three piece round open top metal cans for foods and beverages — Specification
IS 16072 : 2012	Determination of moisture content in milk powder and similar products (routine method)	IS 19396 : 2025/ ISO 23318 : 2022	Milk, dried milk products and cream — Determination of fat content — Gravimetric method
IS 16195 : 2014/ ISO/TS 15495 : 2010	Milk, milk products and infant formulae — Guidelines for the quantitative determination of melamine and cyanuric acid by LC-MS/MS	ISO 21543 : 2020	Milk and milk products — Guidelines for the application of near infrared spectrometry
IS 16424 : 2016/ ISO 7251 : 2005	Microbiology of food and animal feeding stuffs —		

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ANNEX B

(Table 2, Note)

SAMPLING PLAN AND SAMPLING GUIDELINES FOR MICROBIOLOGICAL TESTING OF SKIMMED MILK POWDER

B-1 SAMPLING GUIDELINES

The sampling for different microbiological testing parameters proposed in the standards is to be ensured aseptically by a trained person at manufacturing units following guidelines given in IS 11546. The samples shall be stored and transported under appropriate temperature conditions and insulations within 24 h of sampling to accredited laboratory for analysis as per the approved test methods. A large sample size may be drawn (if desired) according to the tests required and the type of product. Preservatives shall not be added to samples intended for microbiological examination. Three sample sets shall be taken from full production batches. Each sample set shall comprise of a minimum of five samples of 100 g each taken randomly from throughout the batch. The samples will be submitted to the laboratory in the original unopened packaging, sealed at the time of sampling maintained in their original physical state. A set of five samples shall be tested from three different accredited laboratories and the final decision shall be drawn based on three test results. There will be no provision for retesting or re-sampling for microbiological testing.

B-2 SAMPLING PLAN AND INTERPRETATION

The terms n, c, m and M used in this standard have the following meaning:

- n = Number of units comprising a sample.
- c = Maximum allowable number of units having microbiological counts above m for 2-class sampling plan and between m and M for 3-class sampling plan.
- m = Microbiological limit that separates unsatisfactory from satisfactory in a 2-extra class sampling plan or acceptable from satisfactory in a 3-class sampling plan.
- M = Microbiological limit that separates unsatisfactory from satisfactory in a 3-class sampling plan.

B-3 INTERPRETATION OF RESULTS

2-Class Sampling Plan (where n, c and m are specified)	3-Class Sampling Plan (where n, c, m and M are specified)
1) Satisfactory, if all the values observed are $\leq m$	1) Satisfactory, if all the values observed are $\leq m$
2) Unsatisfactory, if one or more of the values observed are $> m$ or more than c values are $> m$	2) Acceptable, if a maximum of c values are between m and M and the rest of the values are observed as $\leq m$
	3) Unsatisfactory, if one or more of the values observed are $> M$ or more than c values are $> m$

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ANNEX C

(Foreword)

COMMITTEE COMPOSITION

Dairy Products Sectional Committee, FAD 19

<i>Organization</i>	<i>Representative(s)</i>
National Dairy Research Institute, Karnal	DR DHEER SINGH (Chairperson)
All India Food Processors Association, New Delhi	SHRI VIJAY GAUR SHRI KUMAR KALPAM (<i>Alternate</i>)
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DR BHAWANA
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Amendments Issued Since Publication

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